Topology: MAT307/M405/MAT803 : - Problem Set 1

1. Let A be a set and $\mathscr{P}(A)$ be its power set. Prove that there is no injective map $f : \mathscr{P}(A) \to A$ and no surjective map $g : A \to \mathscr{P}(A)$.

2. Let X be an infinite set. Is the collection $\tau_\infty := \{U \subseteq X \mid X \setminus U \text{ is infinite or } U = \emptyset\}$ a topology on X? Justify your answer.

3. Let $X = \{a, b, c\}$. Determine which of the following collections of subsets form a valid topology on X. For those that fail, explicitly state which axiom or axioms are violated:
   (i) $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$;
   (ii) $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$;
   (iii) $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$.

4. Let X be a topological space and $A \subset X$. Prove that A is open in X if and only if for each $x \in A$, $\exists$ an open set U in X such that $x \in U \subseteq A$.

5. Let U be an open set and V be a closed set in X. Prove that $U \setminus V$ is open in X and $V \setminus U$ is closed in X.

6. Let $X = \mathbb{N}$ and define $\mathcal{T}_1, \mathcal{T}_2 \subseteq \mathscr{P}(\mathbb{N})$ as follows:
   (i) $\mathcal{T}_1 := \{\emptyset, \mathbb{N}, \{1\}, \{1, 2\}, \{1, 2, 3\}, \{1, 2, 3, 4\}, \dots\}$.
   (ii) $\mathcal{T}_2 := \{\emptyset, \mathbb{N}, \mathbb{N} \setminus 1, \mathbb{N} \setminus \{1, 2\}, \mathbb{N} \setminus \{1, 2, 3\}, \mathbb{N} \setminus \{1, 2, 3, 4\}, \dots\}$.
   Determine whether $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

7. Show that if $\mathscr{B}$ is a basis for a topology on X, then the topology generated by $\mathscr{B}$ equals the intersection of all topologies on X that contain $\mathscr{B}$.

8. Let X be an infinite set and $\mathcal{T}_f$ be the finite complement topology on X. If U and V are non-empty open sets in $(X,\ \mathcal{T}_f)$, prove that $U \cap V \neq \emptyset$.

9. Let $X \neq \emptyset$ be set and let $a \in X$ be a fixed point. Prove that the particular point collection
$$\mathcal{T}_a = \{V \subseteq X \mid V = \emptyset \text{ or } a \in V\}$$
defines a valid topology on X.

10. Let $X \neq \emptyset$ be set and let $a \in X$ be a fixed point. Prove that the excluded point collection
$$\mathcal{T}^a := \{V \subseteq X \mid V = X \ \text{ or } \ a \notin V\}$$
defines a valid topology on X.

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11. Let X be an infinite set and let $a \in X$ be a fixed point. Define a collection of subsets $\mathcal{T}_a \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T}_a = \mathcal{P}(X \setminus \{a\}) \cup \{V \subseteq X \mid a \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Prove that $\mathcal{T}_a$ defines a valid topology on X. What happens when X is a finite set ?

12. Let X be an infinite set and let $a, b \in X$ be two distinct points. Define a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T} = \mathcal{P}(X \setminus \{a, b\}) \cup \{V \subseteq X \mid a \in V \text{ or } b \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Is $\mathcal{T}$ a topology on X?

13. Let $\mathscr{B}_1$, $\mathscr{B}_2$, and $\mathscr{B}_3$ be respectively the collections of all open balls, open rectangles, and open triangles in the plane $\mathbb{R}^2$.
    (i). Prove that all three collections satisfy the basis axioms on $\mathbb{R}^2$.
    (ii). Let $\mathcal{T}_i$ be the topology generated by the basis $\mathscr{B}_i$ for $i = 1, 2, 3$. Let $f : (\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$ be the identity map. Prove that for any choice of $i, j \in \{1, 2, 3\}$, the map f is continuous.
    (iii). Conclude that $\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3$.

14. Along with the finite complement topology ($\mathcal{T}_f$) and the countable complement topology ($\mathcal{T}_c$), we consider the topologies generated by the following six bases:
    (i) **Standard Topology ($\mathcal{T}_{\text{std}}$)**: Generated by the basis of all open intervals:
    $$\mathscr{B}_1 = \{(a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$
    (ii) **K-Topology ($\mathcal{T}_K$)**: Generated by the basis:
    $$\mathscr{B}_2 = \{(a, b) \mid a < b\} \cup \{(a, b) \setminus K \mid a < b\}, \text{ where } K = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}.$$
    (iii) **Lower Limit Topology ($\mathcal{T}_\ell$)**: Generated by the basis of all half-open intervals closed on the left:
    $$\mathscr{B}_3 = \{[a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$
    (iv) **Upper Limit Topology ($\mathcal{T}_u$)**: Generated by the basis of all half-open intervals closed on the right:
    $$\mathscr{B}_4 = \{(a, b] \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$
    (v) **Left-Ray Topology ($\mathcal{T}_{\text{left}}$)**: Generated by the basis of open left rays:
    $$\mathscr{B}_5 = \{(-\infty, b) \mid b \in \mathbb{R}\}.$$
    (vi) **Right-Ray Topology ($\mathcal{T}_{\text{right}}$)**: Generated by the basis of open right rays:
    $$\mathscr{B}_6 = \{(a, \infty) \mid a \in \mathbb{R}\}.$$

    For each pair of these topologies, determine whether one contains the other (i.e., whether one is finer, coarser, or if they are completely incomparable). Summarize your results by drawing a partial order hierarchy diagram representing the containment relations among all eight topologies.

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15. Let $\mathscr{T}_X$ denote the collection of all topologies on a non-empty set X and let $\mathscr{T}_X/\text{hom}$ the set of homeomorphism classes of topologies on X. Let S_X be the group of all bijective maps on X. Note that each bijection $f : X \to X$ induces a bijection $\tilde{f} : \mathscr{P}(X) \to \mathscr{P}(X)$ defined by $\tilde{f}(A) = \{f(a) \mid a \in A\}$ for every subset $A \in \mathscr{P}(X)$. Consequently, S_X acts naturally on $\mathscr{T}_X$ via $f.\mathcal{T} = \tilde{f}(\mathcal{T})$ for all $f \in S_X$, and $\mathcal{T} \in \mathscr{T}_X$. Prove that there is a natural bijection between $\mathscr{T}_X/\text{hom}$ and the collection of orbits of the natural action of S_X on $\mathscr{T}_X$.

16. List down all possible (there are 29) topologies on a three-point set $X = \{a, b, c\}$. Compute the number of orbits under the natural action of S_3 on the set $\mathscr{T}_X$ of all topologies on X.

17. Let X be a set, and let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subbases on X generating topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.
    (i). If $\mathcal{S}_1 \subseteq \mathcal{S}_2$, prove that $\mathcal{T}_1 \subseteq \mathcal{T}_2$.
    (ii). Give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

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